

Defensible evidence in the age of generative AI

CX Impact and Behaviours

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CXO | ESDC



How do we trust AI-generated evidence? Lessons from prediction, thematic analysis, and real-world service improvement.

Dr. Cody Dodd

I wear many hats. Today is about blending theory with evidence. I have been at the forefront of asking when and how should AI outputs support real decisions. I am here to bring lessons from those experiences.



Senior Data Scientist

Client Experience Office

Supports data workflows, research automation methodology, and advanced statistical models.

Applied thematic analysis on thousands of real service comments for departments around the world, driving real-time performance.



Academia

London School of Economics

PhD, Management – Information Systems

- Ethnography of AI algorithms
- Ongoing research in AI platform governance
- Adjunct code-camp instructor



Architecture

Past work

Frameworks driving operational service improvements with proven ROI across dozens of government departments around the world.

Frameworks have directly shaped platforms and technologies (like LSE and PigeonLine's ResearchAI)

Service Canada's Client Experience Office

CXO offers **flexible support**, tailored to priorities and available resources – ranging from **rapid diagnostic and advisory work to modernization support**.

Services CXO offers and how they help:



Journey mapping and service blueprints

See the process end-to-end through the client lens: e.g., where clients get stuck, where expectations are unclear, moments generating calls or complaints. Connect the client journey with the internal operations and pinpoint duplication and bottlenecks.



Sludge assessments and behaviourally-informed solutions

Identify unnecessary burden in processes and preferable behaviours, and design communications to encourage adoption. Review processes around how people behave to eliminate error-prone issues and reduce preventable workload. This helps remove friction that creates pressure for both clients and the staff.



Client insights

Generate actionable insights on client experience – and how operations shape it – by combining client-centred evidence (e.g., annual CX Survey, client feedback) with operational data and validating them with operational experts.



Plain language review and usability testing

Test forms, letters, and digital tools to prevent design issues from generating unnecessary demand.



CX impact measurement

Assess whether an initiative led to meaningful improvements in outcomes for clients and the organization by identifying key behavioural outcomes, determining methods of measurement, examining whether existing organizational data can be used for direct or indirect outcome measurement, and supporting the creation of new metrics where needed.

Thesis



I can't simply assume the answer was generated through a reproducible process. Why does that matter?

Our problem is not really AI, it is that we've built governance around assumptions about AI that are not true.

Traditional assumptions about AI as just another software:

Stable rules, models and outputs

Generative systems introduce:

Changing prompts, models and outputs

Framework

Validation-First AI means understanding risks at the algorithmic level and mitigating it with human-driven science.

The risks we know

Phenomenal for:

- Reformatting
- “have you thought of”
- Faster starts
- Faster coding*

But...

The cold reality

1. **95% of AI investments fail to deliver** - [MIT Labs](#) + [HBR](#)
2. **41% of workers encounter workslop in their work regularly** (fake or empty outputs, lost trust in public institutions)
3. **Only 5 out of 45 sources were real** In a major consulting study, ironically on the topic of AI adoption. [Inside KPMG's Pulled AI Report: Fake Case Studies, Hallucinated Citations, and a Broader Consulting Crisis | Answer | Studio Global AI](#)
4. ***Major studies revealing developer spending more time, not less.** [\[2507.09089\] Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity](#)

Why? Non-deterministic.

- It keeps ignoring parts of my request.
- It summed up only parts
- It made that one up!

Context is a goldilocks zone. Too little, it guesses. Too much, it gets confused or take shortcuts.

Activity block 1 - Prediction

PIMA data

Smith, J.W., Everhart, J.E., Dickson, W.C., Knowler, W.C., & Johannes, R.S. (1988).
UCI Machine Learning Repository:
<https://www.kaggle.com/code/rerere/pima-indians-diabetes-database>

Bank data

Moro, S., Cortez, P., & Rita, P. (2014). UCI Machine Learning Repository:
<https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>

Predicting Diabetes (or campaign success)

This activity introduces deterministic machine learning using the PIMA dataset (or bank dataset).

See ‘[predicting_diabetes/machine_learning.py](#)’

What we will do:

- Deterministic ML can produce stable, reproducible predictions
- Compare Python results to an LLM asked to predict diabetes without labels

TASK:

Using an LLM, and one of the above datasets, write a prompt to add a new column to the end of the dataset, and add a 0 or 1 predicting the outcome based on the row data. You can choose to inform the prompt about the source of data, or keep it generic. Output should be xlsx or CSV.

PIMA data

Smith, J.W., Everhart, J.E., Dickson, W.C., Knowler, W.C., & Johannes, R.S. (1988).

UCI Machine Learning Repository:

<https://www.kaggle.com/code/rerere/pima-indians-diabetes-database>

Bank data

Moro, S., Cortez, P., & Rita, P. (2014). UCI Machine Learning Repository:

<https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>

Predicting Diabetes (or campaign success)

“You are a binary classifier. Use which ever model you think is appropriate. For each row of numeric data in the attached csv, add a new column and output 0 or 1. Do not explain. Do not summarize. Do not add commentary. Do not change the format. Export as xlsx.”

What to observe:

- Some limits to reproducibility
- Polluted / contaminated results
 - Some discover the headers and find the answers. Cheating!
- Some LLMs give up. Some share code. Some write code live and compile it live.

Wine preference

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009).
UCI Machine Learning Repository:
<https://archive.ics.uci.edu/ml/datasets/Wine+Quality>

Passenger satisfaction

Invistico Airline Passenger Satisfaction Dataset.
Kaggle Dataset URL:
<https://www.kaggle.com/datasets/teejmaha120/airline-passenger-satisfaction>

Drivers analysis with Random Forests

This activity introduces a more descriptive form of prediction, an exercise designed to stress LLM analysis.

See ``predicting\_drivers/random\_forest.py``

What we will do:

- Start with the Wine Quality dataset, then repeat with the Airline Satisfaction dataset
- Random forests provide ranked feature importance.
- Compare these to LLM-generated “drivers”

TASK:

Using an LLM, and one of the above datasets, write a prompt to identify the outcome variable, then use the remaining columns as predictors for a regression or drivers analysis. The output should be sorted features by feature importance. Output should be a table that includes accuracy.

Wine preference

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009).

UCI Machine Learning Repository:

<https://archive.ics.uci.edu/ml/datasets/Wine+Quality>

Passenger satisfaction

Invistico Airline Passenger Satisfaction Dataset.

Kaggle Dataset URL:

<https://www.kaggle.com/datasets/teejmaha120/airline-passenger-satisfaction>

Drivers analysis with Random Forests

Find an example prompt here: https://github.com/codydodd/validation-first-ai-workshop/blob/main/predicting_drivers/random_forest.py

What to observe:

- Some get close, if they execute real python. Less so when they guess.
- Others hallucinate drivers entirely.
- Some do not bother. Like before, some may at least provide code.

Computation versus simulation

Python runs deterministically to compute results

LLMs use pattern simulation to ‘fill in the blank’

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #1

“An LLM can read a huge dataset better than a human because it processes every row.”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #1

“An LLM can read a huge dataset better than a human because it processes every row.”

✗ False

Reality:

- LLMs operate through tokenization and attention mechanisms.
- Long datasets get compressed, truncated, sampled, ignored, or receive uneven attention.
- Models are not guaranteed to consider every row equally.
- Even when context fits, "being present" in context is not the same as "being analyzed."

Lesson: LLMs skip a lot.

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

“You should keep adding more documents to the same chat because the LLM builds knowledge over time.”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

“You should keep adding more documents to the same chat because the LLM builds knowledge over time.”

✗ False

Reality:

- Context windows fill quickly.
- Larger contexts increase cost, latency, and complexity.
- Retrieval quality tends to degrade.
- Important instructions may become less influential.
- Many AI engineers recommend periodically starting fresh contexts.

Lesson: Bigger context is not always better context.

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #3

“The model tries to please me instead of being accurate or truthful.”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #3

“The model tries to please me instead of being accurate or truthful.”

 **True**

Reality:

- The model is optimized to generate plausible responses based on a prompt, not on data truthfulness.
- Training often rewards helpfulness and agreement.
- Sycophancy is a documented phenomenon.
- Users frequently interpret confidence as correctness.

Lesson: Plausibility and truth are different objectives.

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #4

“If two people ask the same question, they should get the same answer.”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #4

“If two people ask the same question, they should get the same answer.”

✗ False

Reality:

- Different models.
- Different system prompts.
- Different retrieval.
- Different histories.
- Different random sampling.

Even seemingly identical prompts can produce different outputs.

Lesson: Reproducibility must be demonstrated, not assumed.

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #5

“A bigger model is not always better!”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #5

“A bigger model is not always better!”

 **True**

Reality:

- Bigger models are typically more capable.
- Capability and accuracy are not the same thing.
- For narrow classification tasks, a benchmarked classifier often beats a frontier LLM.
- For thematic analysis, a validated taxonomy often matters more than model size.

Lesson: Better governed often beats bigger.

Activity block 2 – Text analysis

Why thematic analysis?

Common approaches

- **Summarization** → At a high level, generally speaking, what is being said?
- **Sentiment analysis** → How do people feel? (what can we do with this data though?)
- **Clustering / Topic Models** → What patterns exist? (these are rarely human interpretable)
- **Thematic analysis** → What should we understand and act upon?

Why thematic analysis?

When done well, thematic analysis is often the most actionable form of text analysis because it connects individual comments to service, policy, communication, and operational issues. It does this through subject-matter experts and their validated themes.

It also allows us to quantify what is being said:

- Staff compliments: 50%
- Wait time complaints: 20%
- Form issues: 8%
- Longitudinal (what has changed over time)
- Drivers (turns out form issues are less common, but more likely to drive down satisfaction)
- Segmentation (who is most impacted by these themes?)
- Impact (hypothesis testing)

Why thematic analysis?

Thematic analysis means adding a tag to each comment so that we can conduct descriptive or inferential analysis.

Row	Response	Score	Sentiment	Created On	Wants to	Needs tax	Amounts	request to	Missing p	Looking fo	System er	Looking fo d
0	existe	0	Neutral	2026-03-01 0:00:00	1	0	0	0	0	0	0	0
1	Addre	0	Neutral	2026-03-01 0:00:00	1	0	0	0	0	0	0	0
2	Where	-0.8731	Very nega	2026-03-01 0:00:00	0	1	0	0	0	0	0	0
3	Supplé	0	Neutral	2026-03-01 0:00:00	0	0	1	0	0	0	0	0
4	This is	-0.6471	Very nega	2026-03-01 0:00:00	1	0	0	1	0	0	0	0
5	je cher	0	Neutral	2026-03-01 0:00:00	0	1	0	0	0	0	0	0
6	Pas en	0	Neutral	2026-03-01 0:00:00	0	0	0	0	1	0	0	0
7	My hu	0	Neutral	2026-03-01 0:00:00	0	0	1	0	0	0	0	0
8	Amenc	0	Neutral	2026-03-01 0:00:00	0	1	0	0	0	0	0	0
9	old age	0	Neutral	2026-03-01 0:00:00	0	0	0	0	0	1	0	0
10	Applyi	-0.7511	Very nega	2026-03-01 0:00:00	0	0	0	0	0	0	1	0
11	I can't	-0.6708	Very nega	2026-03-02 0:00:00	0	0	0	0	0	0	0	1
12	how i c	0	Neutral	2026-03-02 0:00:00	0	0	1	0	0	0	0	0
13	How d	0	Neutral	2026-03-02 0:00:00	0	0	0	0	0	0	0	0
14	I am lo	0	Neutral	2026-03-02 0:00:00	0	0	0	0	0	1	0	0
15	Can't a	-0.6617	Very nega	2026-03-02 0:00:00	0	0	0	0	0	0	1	0

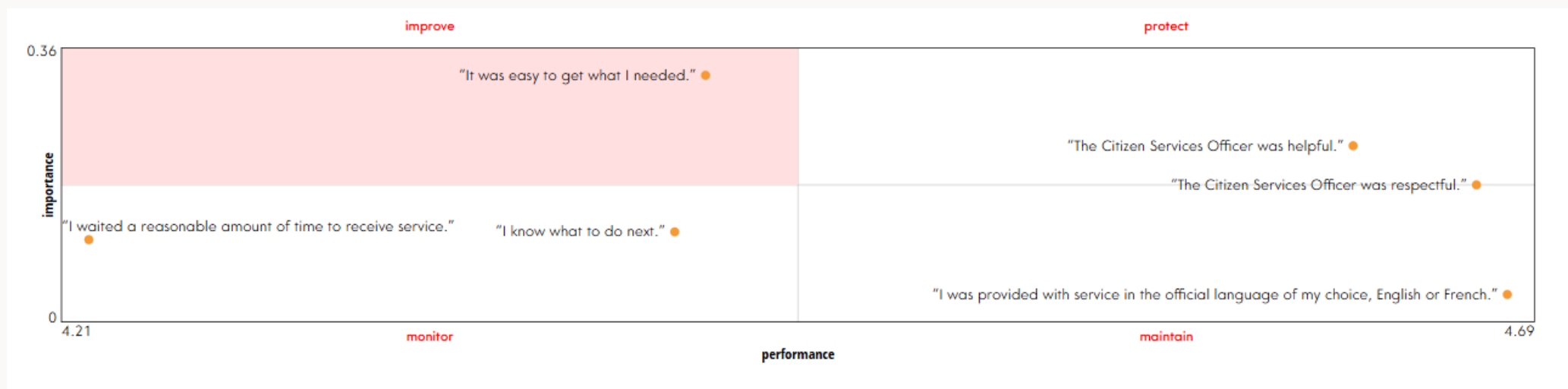
For the data nerds: this is a data matrix! 🥰

Why thematic analysis?

Enables drivers analysis:

Such as seeing which themes correlate or drive outcomes like satisfaction (if comments exist alongside satisfaction scores).

Here themes around 'ease' were more correlated with satisfaction / dissatisfaction than wait times, despite wait times having higher coverage.



Why thematic analysis?

Enables segmentation analysis:

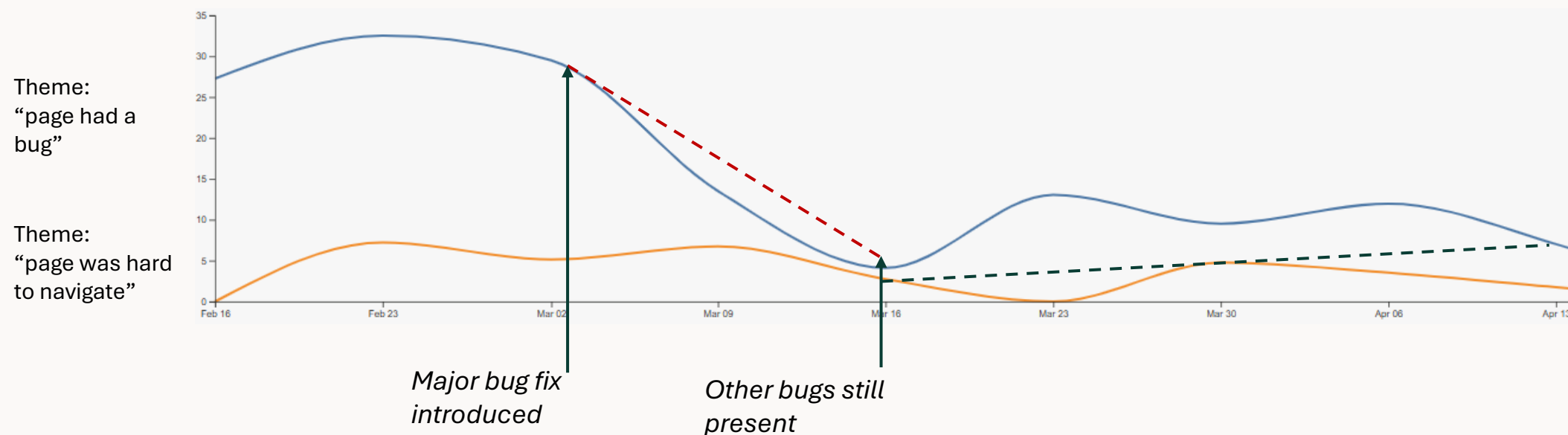
Which categories or operational data are most over- or under- represented by some themes?

Here those over 60 are statistically more likely to compliment staff, while those between 18 and 30 are far less.

Segments/outliers for the theme “staff compliment” compared to the average of 9.4%

FIELD	VALUE	SIZE	COVERAGE	DIFF	SCORES	P-VALUE	CONFIDENCE	INDICATORS
What is your age group?	Aged 60 and over	252	14.3%	4.85	14.57	0.00	10% – 19%	Significant High confidence
What is your age group?	Aged 18 to 30	73	2.7%	-6.70	8.42	0.04	0% – 6%	Significant High confidence
What is your gender?	Male	368	11.1%	1.71	6.61	0.13	8% – 14%	High confidence Not significant
What is your age group?	Aged 31 to 59	472	8.1%	-1.39	6.34	0.11	6% – 11%	High confidence Not significant
What is your gender?	Female	393	8.1%	-1.29	5.24	0.22	5% – 11%	High confidence Not significant
What is your gender?	Prefer not to answer	10	0.0%	-9.44	2.45	0.30	0% – 0%	High confidence Not significant
What is your gender?	Another gender	7	14.3%	4.85	0.91	0.66	0% – 40%	Not significant

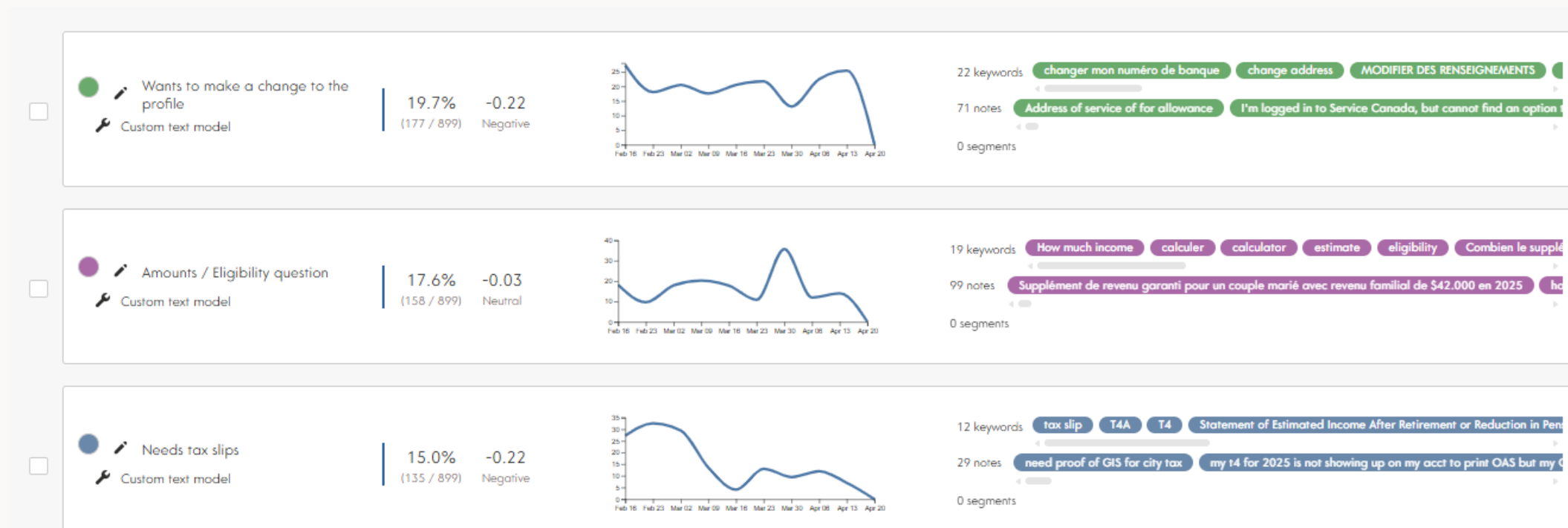
Why thematic analysis?



Stable text themes over time can enable impact measurement. In this case, a major bug was eclipsing other issues like page navigation. A fix was implemented, and “bugs” became a much smaller issue than prior.

Quantifying text is powerful, but only so long as we trust the validation process.

Why thematic analysis?



Example top 3 themes from OAS page feedback, after human-validated tagging. Combines:

- Keywords from analyst-curated dictionary
- Researcher 'notes' can add tags that do not get captured by keywords (and keywords can be dropped)
- Machine-learning matches can be 'corrected' through human input

Accurately tagged comments enables powerful insights measurement.

The challenge becomes validating these tags as all this power depends on it.

Validation-first thematic measurement

Something I have been writing about for some time: [codydodd/validation-first-thematic-measurement](https://codydodd.com/validation-first-thematic-measurement)

Research around knowing ‘When to use what’?

- Human tagging
- Taxonomies / dictionaries (keyword matches)
- Machine learning from human matching
- LLM-tagging ← (not the same as LLM summarization)

But also recognizing the limits of LLM-only analysis:

LLM-only analysis:

Comments → LLM → Themes → Decisions. If audited, how can our decisions and themes be validated?

Validation-first analysis:

Human themes → Human-tagged benchmarks → Automation of dictionaries → Validation / Review →
Decision support

Citizen satisfaction survey – Benefits

Primary source: ICCS Citizen First 2023, benefits programs subset - Combines sources from multiple levels of government to help anonymize the material.

https://github.com/codydodd/validation-first-ai-workshop/blob/PublicAffairs-ESDC/predicting_themes/datasets/CF2023-subset-benefits-anonymized-wave-1-no-results.xlsx

LLM-based Thematic Analysis

This activity introduces a guided approach to using LLMs to running thematic analysis.

TASK:

Version 1:

Using an LLM, and one of the above datasets, write a prompt to identify discrete, actionable themes, with short, clear labels. Sort by most common themes.

Version 2:

Using an LLM, and one of the above datasets, write a prompt to identify discrete, actionable themes, with short, clear labels. Do so for each row (comment) and create a new column for each theme and add a 1 per row where this theme was found. Export as a xlsx.

Citizen satisfaction survey – Benefits

Primary source: ICCS Citizen First 2023, benefits programs subset - Combines sources from multiple levels of government to help anonymize the material.

https://github.com/codydodd/validation-first-ai-workshop/blob/PublicAffairs-ESDC/predicting_themes/datasets/CF2023-subset-benefits-anonymized-wave-1-no-results.xlsx

LLM-based Thematic Analysis

This activity introduces deterministic machine learning using the PIMA dataset.

What to observe:

- Brittle models
- Deterministic stability versus difficult to manage, less accurate
- Generic ML models struggle. Hand-tuned ones can out-perform basic models after time and human management.
- LLMs are inconsistent without python guidance and pushing towards a method that ‘reveals its logic’ through tagging and through an exportable file.

Challenge:

- Won’t do this for tens of thousands of comments very affordably.

Citizen satisfaction survey – Benefits

Primary source: ICCS Citizen First 2023, benefits programs subset - Combines sources from multiple levels of government to help anonymize the material.

https://github.com/codydodd/validation-first-ai-workshop/blob/PublicAffairs-ESDC/predicting_themes/datasets/CF2023-subset-benefits-anonymized-wave-1-no-results.xlsx

Model comparison exercise

A powerful tool for validating models in a robust way, that does not require major commitment:

See `predicting\_themes.py` in the repository.

1. We take a human-validated dataset as a ‘source of truth’
 - Say a representative subset of comments
2. We compare this to:
 - a) analyst-managed keyword dictionaries (do pretty good)
 - b) Naïve machine learning (does not so good!)
 - c) Tuned machine learning (with work gets better than keywords)
 - d) LLM-tuned - decent but requires:
 - Good analyst-tuned models
 - Human in the loop
 - Row-based analysis

This exercise becomes a validation exercise. Can be redone as models need improvement.

Validation-first thematic measurement

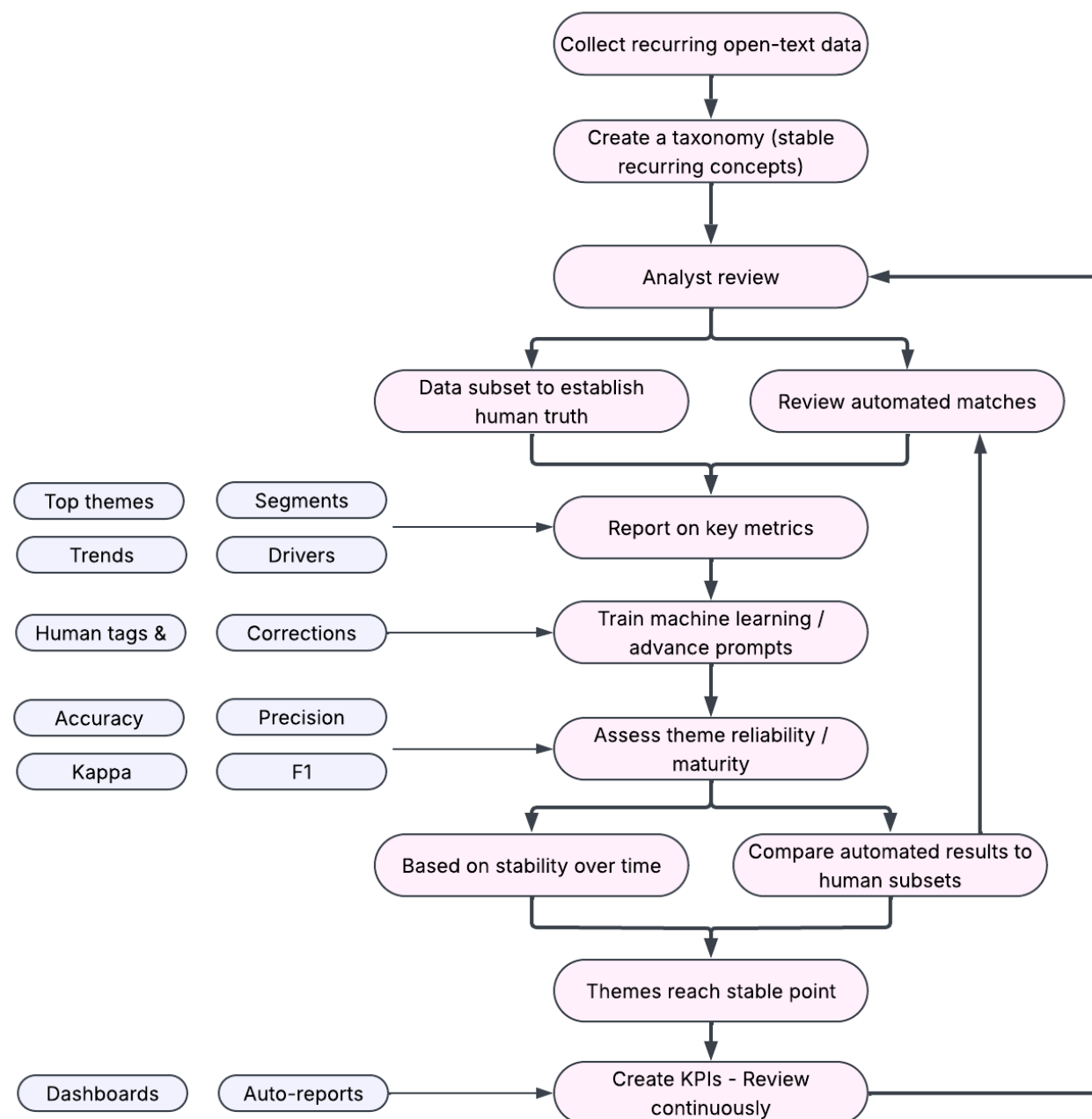
[codydodd/validation-first-thematic-measurement](https://codydodd.com/validation-first-thematic-measurement)

Validation-First Thematic Measurement is a methodology for transforming recurring open-text feedback into trusted, measurable evidence for decision-making.

Core principles

- Human judgement remains the reference standard
- Validation comes before measurement
- Themes must be actionable and operational
- Automation supports analysts, not replaces them
- Only validated classifications should drive decisions

Example Validation-First Thematic Measurement workflow



Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #6

“A summary is evidence.”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #6

“A summary is evidence.”

✗ False

Reality:

Summarization amplifies the goal of targeted attention, which amplifies missing data, sycophancy, and filling in information. Summaries can:

- Omit minority views
- Exaggerate common views
- Introduce interpretation, and hide when it does this
- Hide uncertainty

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #7

“If ten people use AI and get similar answers, the answer is probably correct.”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #7

“If ten people use AI and get similar answers, the answer is probably correct.”

✗ False

All ten systems may:

- have similar training data
- share assumptions
- reproduce the same bias

Consensus can emerge from the model ecosystem rather than from reality.

Lesson: Agreement is not validation.

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #8

“If important information is missing, the LLM will tell me it doesn't know.”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #8

“If important information is missing, the LLM will tell me it doesn't know.”

 False

Reality

LLMs are prediction systems.

When information is missing, they frequently:

- infer
- interpolate
- generalize
- assume
- complete patterns

As uncertainty increases, humans often become less confident.

LLMs frequently remain highly confident because they are trained to continue generating the most plausible next tokens.

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #9

“The hardest part of thematic analysis is finding themes.”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #9

“The hardest part of thematic analysis is finding themes.”

✗ False (in our experience)

The hardest part is:

- creating a defensible taxonomy
- validating results
- maintaining consistency
- linking findings to outcomes

Lesson: Generating themes is often the easy part. Governance is harder than generation.

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #10

“Automation reduces the importance of human expertise.”

Common AI beliefs

Many AI failures happen because we assume things about LLMs that simply aren't true.

True or False #10

“Automation reduces the importance of human expertise.”

✗ False

Your entire workshop basically proves the opposite.

Good AI workflows usually require:

- better subject matter experts
- stronger validation
- better measurement
- clearer governance

not less.

Lesson: Human expertise becomes more important as systems become more generative.

AI governance

Every one of these misconceptions comes from treating LLMs as if they were deterministic software. They're not.

That's why validation becomes the central governance challenge.

Frameworks like VFTM are not about 'ways of doing thematic analysis', but about tackling broader governance problems. In the research world, we are accountable to the insights our tool-use produces.

Thank you.

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